

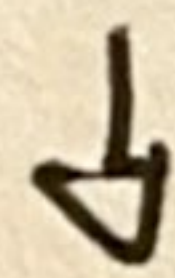
Main Question: What is algebra?

If you ask someone on the street this question, the most likely response will be

"something horrible to do with x, y and z "

What if you asked a mathematician instead?

"Algebra is the abstract encapsulation for our intuition for composition"



By this we mean: The concept of 2 objects put together to form a new one.

Example: Adding two numbers

- composing real valued single variable functions.

We'll see together the simple idea of composition hides vast hidden depth.

- Algebra permeates all our mathematical intuitions.
- In fact, the first mathematical concepts we encountered are the foundation of the subject.

↳ Let's try to summarize some of them!
(in your previous years)

The concept of unity:
The number 1.



$\mathbb{N} = \{1, 2, 3, \dots\}$ the natural numbers.

$\mathbb{N}$ we equipped with 2 operations

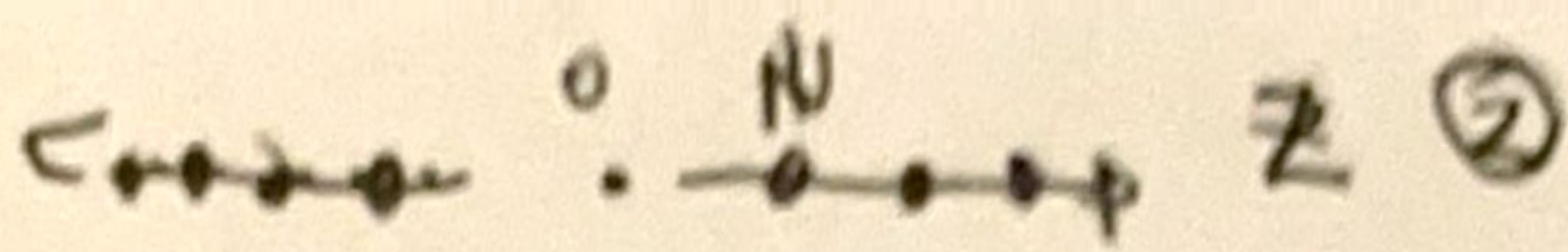
- (product) and + (sum)



$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ The integers.

We formalize by using geometric intuition thinking of $\mathbb{N}$ as sitting on

a line.



$\mathbb{Z}$ also comes with 2 operations:

- $(\mathbb{Z}, +)$ $\rightarrow$ has a nice property now: we have additive inverses!
(compare to $(\mathbb{N}, +)$)
- $(\mathbb{Z}, \cdot)$ $\rightarrow$ we don't have inverses.



the rationals: $\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$

we form these by taking $\mathbb{Z}$ and formally dividing through by non-negative integers.

Again we have 2 operations:

- $(\mathbb{Q}, +)$
- $(\mathbb{Q}^{\times}, \cdot)$ $\rightarrow$ this time multiplication has particularly good properties e.g. non-zero elements have an inverse.

We could keep going! by forming

$\mathbb{R}$ the set of real numbers

$\mathbb{C}$ the set of complex numbers

The process of constructing them is slightly more ~~com~~ intricate,

• $\rightarrow$ for $\mathbb{R} \rightarrow$ goes to the realm of mathematical ~~anaylsis~~ analysis.

• $\rightarrow$ for $\mathbb{C} \rightarrow$ goes to the second part of the series of an algebra course

Key point! Notice that at each stage the operations $+$ and $\cdot$ gain additional properties.

In high school you spend reasonable amount of time understanding how $+$ and $\cdot$ behave in $\mathbb{Q}$

For example:

③

$$\bullet a+b=b+a \quad \forall a, b \in \mathbb{R}$$

$$\bullet a \cdot (b+c) = a \cdot b + a \cdot c \quad \forall a, b, c \in \mathbb{R}$$

The central idea behind abstract algebra is to define a larger class of objects (sets with extra structure) of which ~~$(\mathbb{Z}, +)$~~ and ~~$(\mathbb{R}, \cdot)$~~

$\mathbb{Z}$ and $\mathbb{Q}$ are definitive members:

$$\begin{array}{l} \bullet (\mathbb{Z}, +) \rightarrow \underline{\text{Groups}} \\ \bullet (\mathbb{Q}^\times, \cdot), (\mathbb{Q}, +) \end{array} \left. \vphantom{\begin{array}{l} \bullet (\mathbb{Z}, +) \rightarrow \underline{\text{Groups}} \\ \bullet (\mathbb{Q}^\times, \cdot), (\mathbb{Q}, +) \end{array}} \right] \leftarrow \begin{array}{l} \text{Our subject} \\ \text{of study in} \\ \text{This class} \end{array}$$

Not example of a group: $(\mathbb{N}, +)$.

Later: $(\mathbb{Z}, +, \cdot) \rightarrow \text{ring}$
 $(\mathbb{Q}, +, \cdot) \rightarrow \text{field}$ } PART 2.
OF ABSTRACT ALGEBRA.

In linear algebra you did something similar:

nice example:

$(\mathbb{R}^n, +, \text{scalar multiplication})$ ^{Abstract} $\longrightarrow$ Vector spaces over $\mathbb{R}$.

surprising factor: These vague ideas mean something very precise and have far more depth than one could ever imagine.

Lecture 1: SETS AND FUNCTIONS.

learning Goals of The ^{week} day:

- 1) state the definition of a set and give examples
- 2) learn how to handle operations of sets: union, intersection, cartesian product.
- 3) Definition of a function and how to prove it satisfies a property: injective, surjective, bijective.

4) Equivalence relations: state their definition and how to prove/disprove a relation is an e.r.

Q: what is a set?

Definition: A set is any collection of objects.

Examples: 1) Six dogs

2) all protons on Earth

3) every though you have ever had

4) $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Note that past time we distinguished 2

operations on them $\left. \begin{array}{l} + \text{ sum} \\ \cdot \text{ product.} \end{array} \right\} \rightarrow \text{on } \mathbb{Z} \text{ and } \mathbb{Q}$

so we considered $\mathbb{Q}$ and $\mathbb{Z}$ with some extra structure: the sets with these operations.

In this course: we will study sets with some carefully chosen extra structure.

NOTATION (LOGIC)

a) If P and Q are 2 statements then

$P \rightarrow Q$ means if P is true then Q is true.

e.g. $x \text{ is odd} \Rightarrow x \neq 2$

We say that P implies Q .

b) If $P \rightarrow Q$ and $Q \rightarrow P$ then we

write $P \leftrightarrow Q$ which should be read as P is true ~~iff~~ if and only if Q is true.

c) The symbol $\forall$ should be read as "for all"

d) The symbol $\exists$ should be read as "there exists".

$\exists!$ should be read as "There is a unique"

NOTATION : (SETS)

②

Let S and T be two sets:

a) if s is an object in S then we say that s is an element or member of S .

We write $s \in S$ to express this.

~~eg:~~ and $s \notin S$ to say " s is NOT an element in S ".

e.g. : $5 \in \mathbb{R}$

$\frac{1}{2} \notin \mathbb{Z}$

b) The standard way ^{of} ~~to~~ writing down a set S is using curly brackets:

$$S = \{ \text{Notation for element in } S \mid \text{Properties which specify being in } S \}$$

The vertical line should be read as "such that".

examples:

- S is the set of all even integers:

$$S = \{x \in \mathbb{Z} \mid 2 \text{ divides } x\}.$$

- $S = \{1, 2, 3\}$ for S finite we just list the elements.

c) if every object of S is also an element of T then we say that S is contained in T .

we write $S \subset T$ to denote this.

to say S is NOT contained in T we write $S \not\subset T$.

e.g.:

• $\mathbb{N} \subset \mathbb{Z}$		$\mathbb{Q} \not\subset \mathbb{N}$
• $\mathbb{Z} \subset \mathbb{Q}$		

Remark: if $S \subset T$ and $T \subset S$ Then $S = T$.

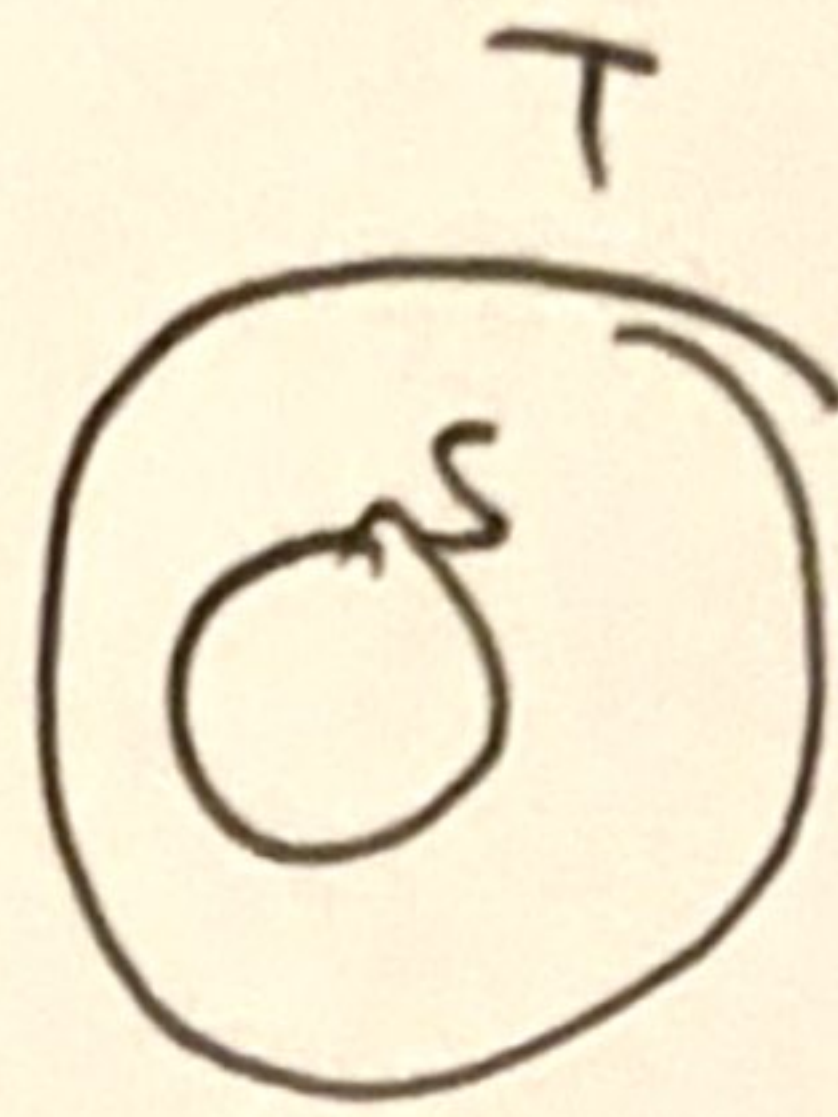
we write $S \subseteq T$: to say " S is contained in T or is equal to T ".

e.g.: $\mathbb{Z} \subseteq \mathbb{Z}$ $\mathbb{N} \subseteq \mathbb{Z}$

a) If $S \subsetneq T$: "S is contained in T (3)
but NOT equal to T".

e.g.: $\mathbb{N} \subsetneq \mathbb{Z}$

and we say S is a
proper subset of T.



b) If $S \subset T$ then $T \setminus S = \{x \in T \mid x \notin S\}$

e.g. $\mathbb{N} \subset \mathbb{Z}$ $\mathbb{Z} \setminus \mathbb{N} = \{-1, -2, -3, \dots\}$

and $T \setminus S$ is called the complement
of S in T.

c) The set of elements contained in both
S and T is called the intersection
of S and T.

We write this as $S \cap T$.

f) The collection of all objects which are either in S or T is called the union of S and T .

We write $S \cup T$ to denote this.

g) $S \times T = \{ (a, b) \mid a \in S, b \in T \}$ We call this new set the cartesian product of S and T . We might naturally extend this concept to finite collection of sets.

e.g. $\mathbb{R}^2 = \{ (x, y) \mid x \in \mathbb{R}, y \in \mathbb{R} \}$

$$\mathbb{R} \times \mathbb{N} = \{ (x, n) \mid x \in \mathbb{R}, n \in \mathbb{N} \}$$

n) The set which contains no objects is $\emptyset$ called the empty set. We write $\emptyset$ to denote this.

i) we say that 2 sets are disjoint if $S \cap T = \emptyset$.

The union of two disjoint sets is often written as $S \cup T$.

e.g. $(0, 1)$ $(2, 3)$
disjoint

$(2, 3)$ $(2, 4)$
 $\begin{array}{cccc} 1 & 2 & 3 & 4 \\ | & | & | & | \\ + & - & - & - \end{array}$
($(2, 4)$ $\uparrow$
not disjoint.

Definition: A map (of function) f from S to T is a rule that assigns to each element of S a unique element of T .

We express this information using the following notation:

$$\begin{aligned} f: S &\longrightarrow T \\ x &\longrightarrow f(x) \end{aligned}$$

Examples:

$$\begin{aligned} 1. \quad S = T = \mathbb{N} \quad & f: \mathbb{N} \longrightarrow \mathbb{N} \\ & a \longrightarrow a^2. \end{aligned}$$

$$2. \quad S = \mathbb{Z} \times \mathbb{Z} \quad T = \mathbb{Z}$$

$$\begin{aligned} f: \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (a, b) &\longrightarrow a + b \end{aligned}$$

This concept is simple! but it unravels enormous depth! to illustrate this observe that calculus is the study of certain classes of functions (continuous, differentiable or integrable) from $\mathbb{R}$ to $\mathbb{R}$.

Definition 2: Let S and T be two sets, and $f: S \rightarrow T$ be a map. ⑤

1) We say that S is the domain of f and T is the codomain of f .

2) We say that f is the identity map if $S = T$ and $f(x) = x \quad \forall x \in S$.

In this case we write $f = \text{Id}_S$.

3) f is injective if $f(x) = f(y)$ implies that $x = y \quad \forall x, y \in S$.

Example

$$\mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x+1$$

Not example

$$\mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x^2$$

$$(-1)^2 = 1^2 = 1$$

4) f is surjective if given $y \in T$ there is $x \in S$ such that $f(x) = y$.

Example

$$\mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x+1$$

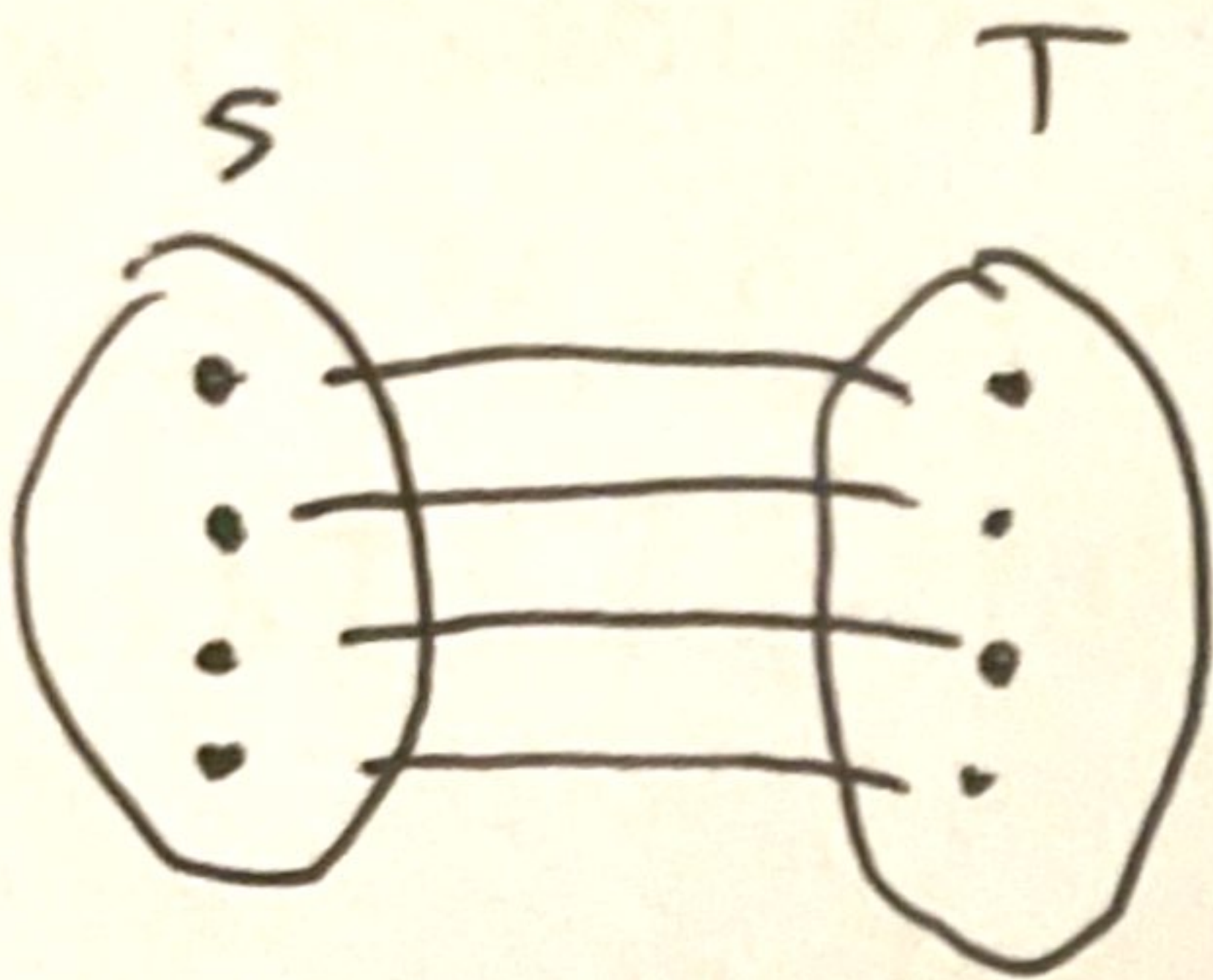
Not example.

$$\mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x^2$$

Observe

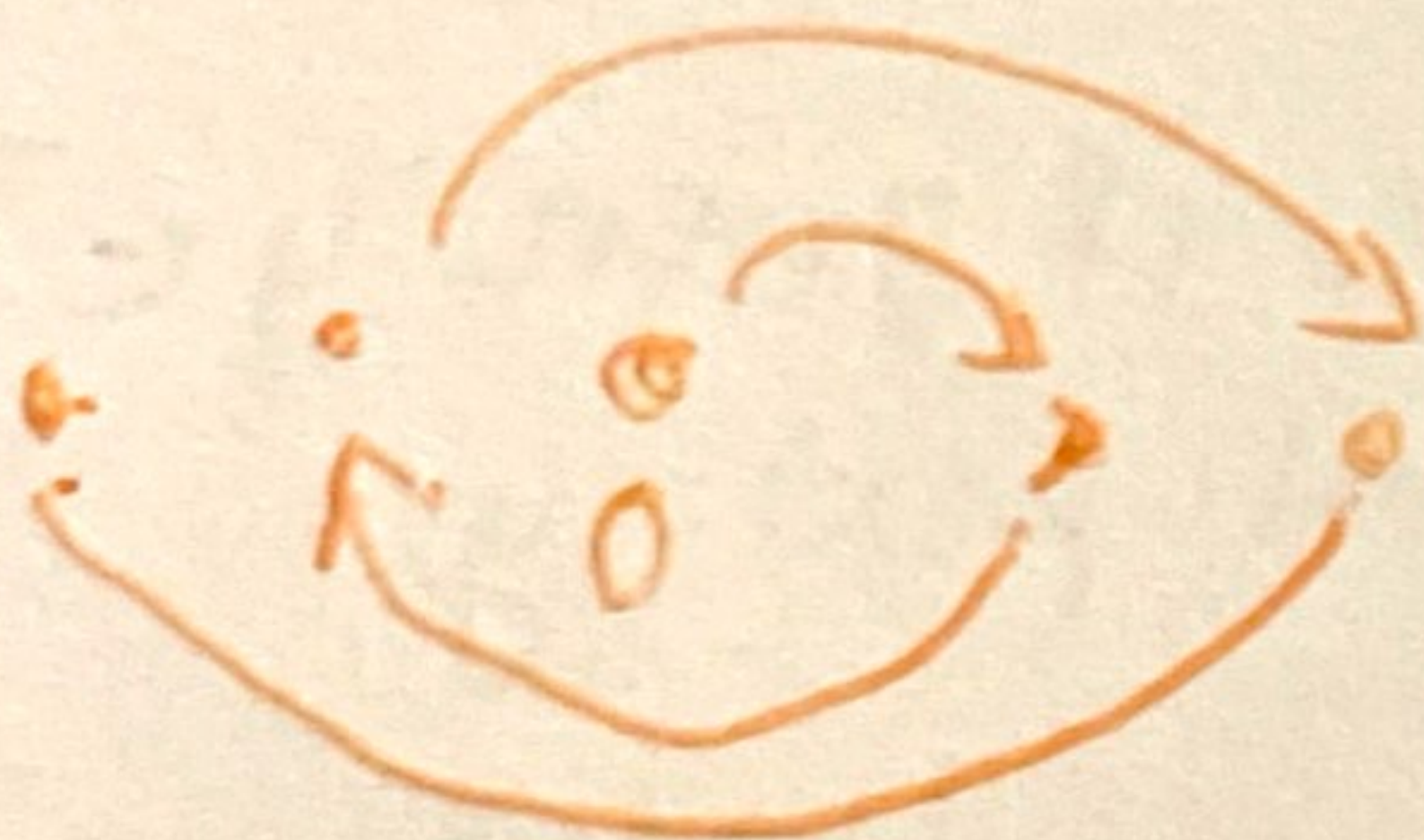
5) If f is both injective and surjective we say that it is bijective. Intuitively this means that there is a perfect matching of elements in S and T .



Examples:

$$4) f: \mathbb{N} \rightarrow \mathbb{Z}$$

Idea: $-2 \ -1 \ 0 \ 1 \ 2$
 $f(4) \ f(2) \ f(0) \ f(1) \ f(3)$



Definition: The size of a set S is the number⁽⁶⁾
(cardinality)

of elements, and we denote it by $|S|$

- $|\{a, b, c\}| = 3$
- $|\mathbb{N}| = \aleph_0$ and $|\mathbb{R}| = 2^{\aleph_0}$ (not the subject of the course)
- Q: When is it the case that two given sets have the same cardinality?

Answer: $|S| = |T|$ iff there is a bijective function $f: S \rightarrow T$.

e.g: $\overset{T}{S} = \left\{ \frac{1}{n} \mid n \in \mathbb{N}^{\geq 1} \right\}$

$$\mathbb{N}^{\geq 1} = S = \{n \in \mathbb{N} \mid n \geq 1\}$$

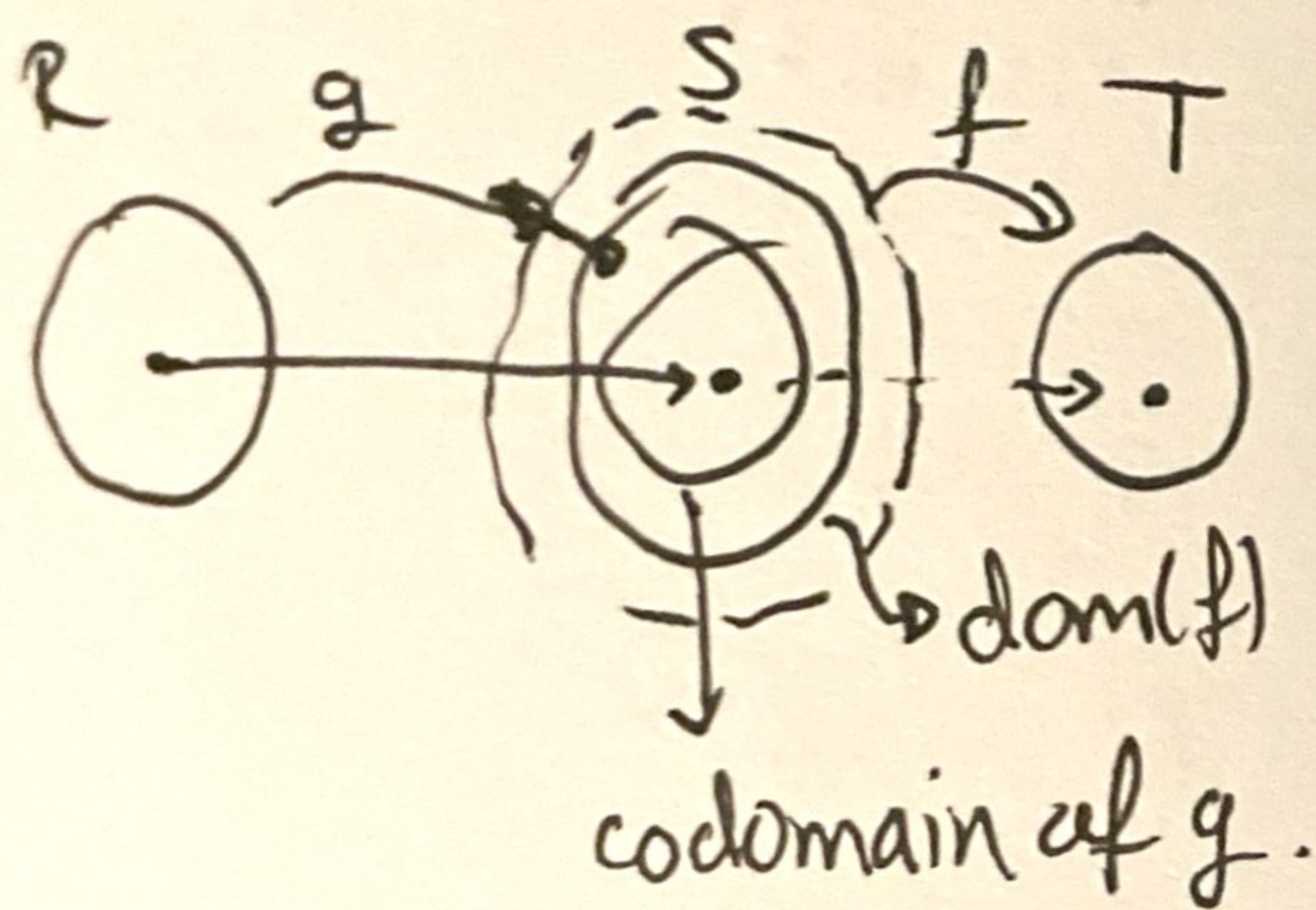
$$f: \mathbb{N} \rightarrow T$$

$$n \rightarrow \frac{1}{n}.$$

observation: If R, S and T are sets and $g: R \rightarrow S$ and $f: S \rightarrow T$ are maps then we may compose them to a new function $f \circ g: R \rightarrow T$

We might compose them to give a new function $f \circ g: R \rightarrow T$.

Warning! Only possible if the domain of f is contained in the codomain of g .



[Ex] Let S and T be two sets. Let f be a map from S to T . Show that f is a bijection iff there exists a map g from T to S such that $f \circ g = id_T$ and $g \circ f = Id_S$.

Equivalence Relations:

Intuition: Within a set it is sometimes natural to talk about different elements being related in some way.

For example, we could say $x, y \in \mathbb{Z}$ are related if $x - y$ is divisible by 2.

Said another way, x and y are related if:

- they are both odd,
- they are both even,

This idea can be formalized as something called AN EQUIVALENCE RELATION.

Definition: An equivalence relation on a set S is a subset $\underline{U} \subseteq S \times S$ satisfying:

- 1) $\forall x \in S \quad (x, x) \in U$ (reflexivity)
- 2) $(x, y) \in U$ implies $(y, x) \in U$ (symmetry)
- 3) Given $x, y, z \in S$
 $(x, y) \in U$ and $(y, z) \in U$ then
 $(x, z) \in U$ (transitivity)

If $U \subseteq S \times S$ is an equivalence relation then we say that $x, y \in S$ are equivalent if and only if $(x, y) \in U$.

It is more convenient to denote this as $x \sim y$ to mean that x and y are equivalent.

Definition: Let $\sim$ be an equivalence relation on a set S . Let $x \in S$ the equivalence class containing x is ^{the} subset

$$[x] := \{y \in S \mid y \sim x\} \subseteq S.$$

Remarks:

1. Notice that ^{+ symmetry} the transitivity properties imply that $y \in [x]$ iff $[x] = [y]$

Hence

Remarks: 1) the reflexivity property implies that $x \in [x]$.

then equivalence classes are NOT empty and their union is S .

2) symmetry and transitivity imply that $y \in [x]$ iff $[y] = [x]$.

Hence two classes are equal or disjoint.

It should be noted that we can represent a given class using any of its members using $[x]$ notation.

Proof:

we show $[y] \subseteq [x]$

let $z \in [y]$ then $z \sim y$, since $y \in [x]$

then $y \sim x$ by transitivity $z \sim x$ so $z \in [x]$.

The converse of $[x] \subseteq [y]$ follows the same argument, using symmetry we get $x \in [y]$.

Examples:

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(x, y) \mapsto x - y.$$

~~1. Let $\sim$ be~~

1. Define the relation on $\mathbb{R}$ by

Given $a, b \in \mathbb{R}$

$$a \sim b \iff a - b \in \mathbb{Z}$$

show that $\sim$ is an equivalence relation

reflexivity: $a - a = 0 \in \mathbb{Z}$

symmetry: $a \sim b \Rightarrow a - b = z$ for $z \in \mathbb{Z}$

but multiplying by -1 $b - a = -z \in \mathbb{Z}$

Then $b \sim a$.

transitivity: $a \sim b \Rightarrow a - b = z \in \mathbb{Z}$

$$b \sim c \quad b - c = w \in \mathbb{Z}$$

$$a - c = (a - b) + (b - c) = z + w \in \mathbb{Z}$$

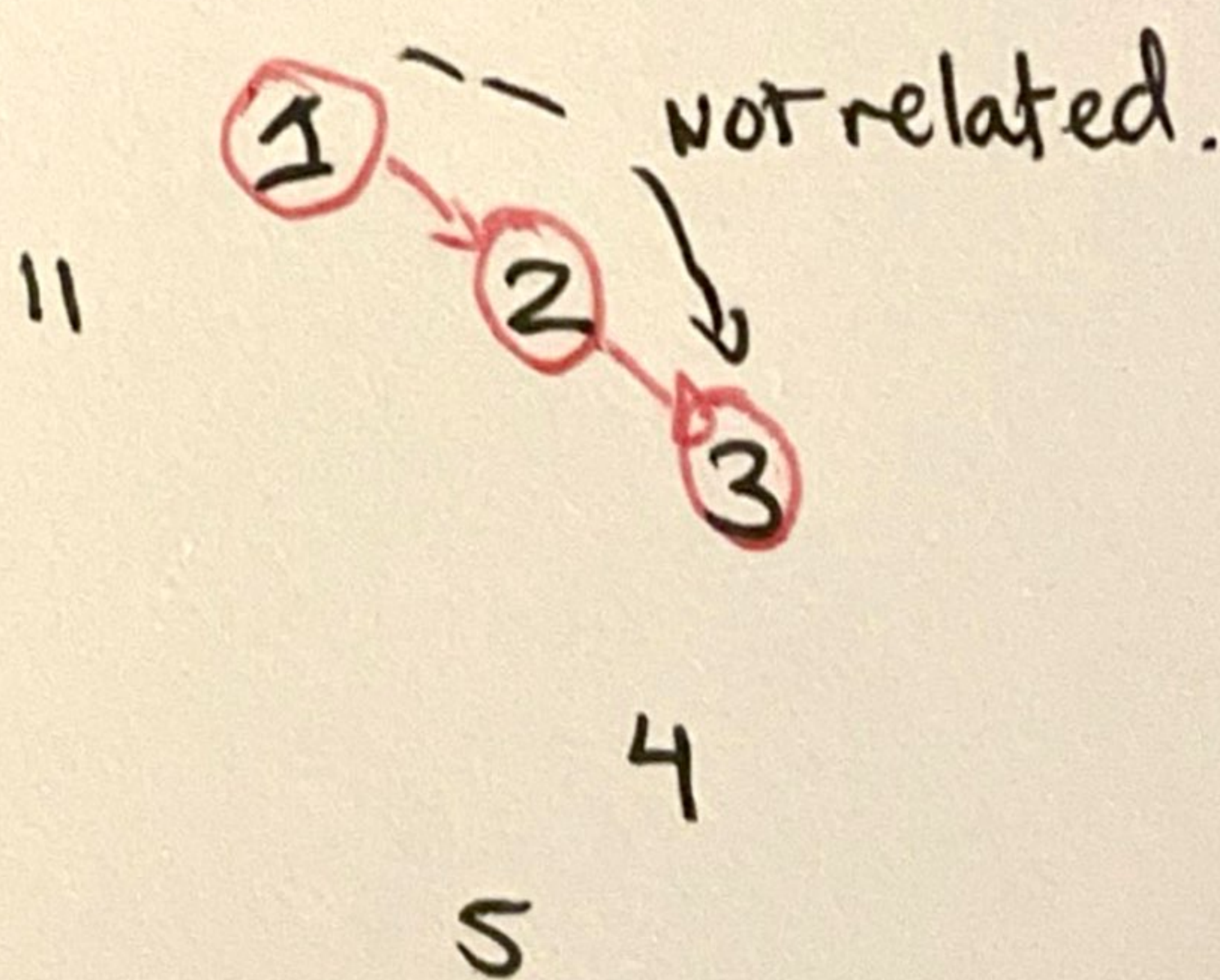
2. let C be the set of number appearing on a standard clock face. For $a, b \in C$ define

$a \sim b \iff b = a$ or b appears next to a on the clock face.

Is this an equivalence relation on C ?

If so prove it if not explain why.

No.



$$A \Delta B = A \cup B - (A \cap B)$$

definition: Let S be a set. Let $\{X_i\}$ be a collection of subsets. We say that $\{X_i\}$ forms a partition of S if each X_i is non empty they are pairwise disjoint and their union is S .

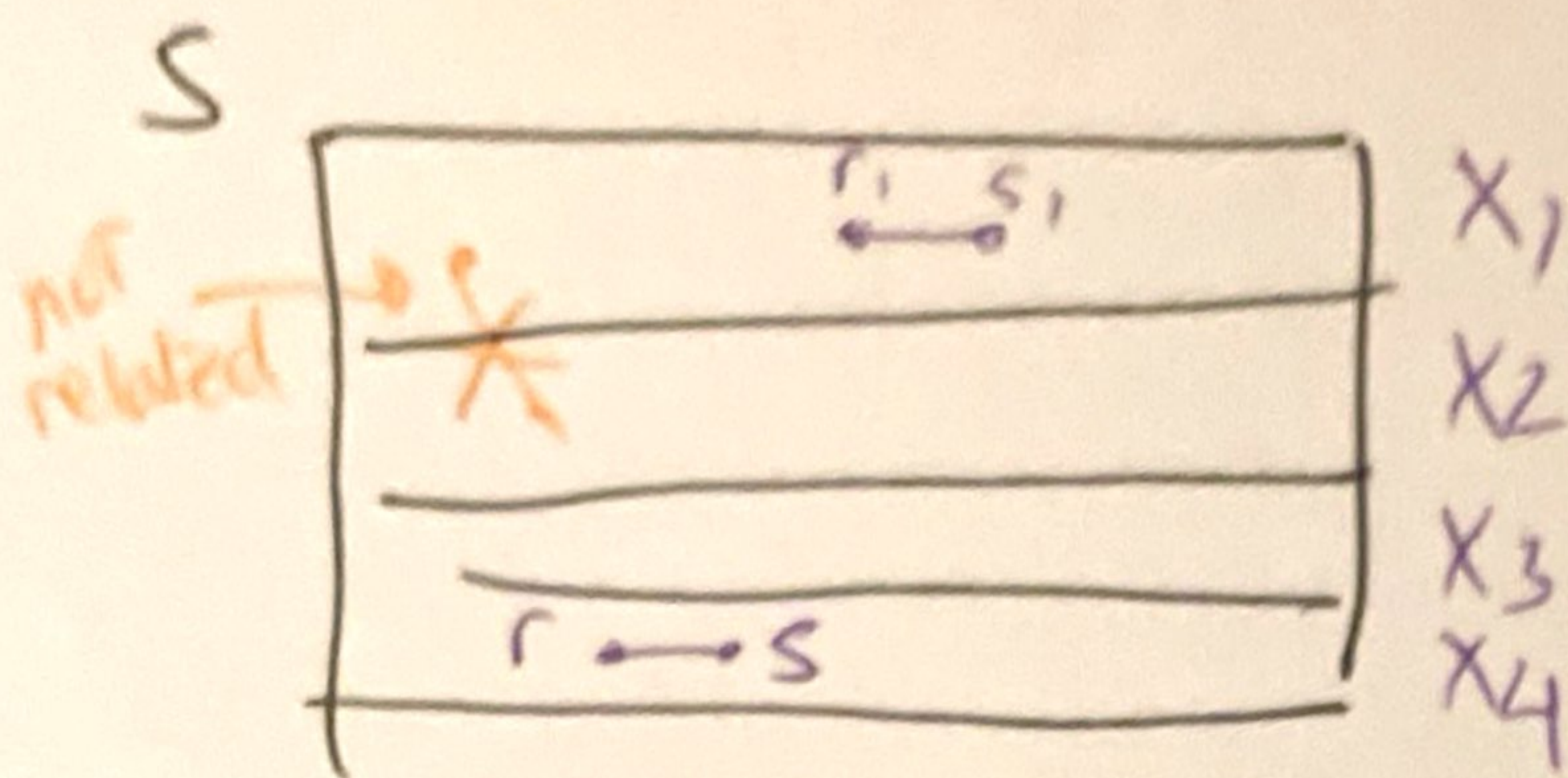
We just showed that any equivalence relation induces a partition on the set!

- The sets X_i are the equivalence classes
- Their union covers S .

The converse is also true! i.e.

Proposition: Let S be a set and $\{X_i\}_{i \in I}$ be a partition of S . Let

$$E = \{(r, s) \in S \times S \mid \exists i \in I \quad r, s \in X_i\}$$



E is an equivalence relation.

Proof: reflexivity: $r \in X_i$ for some i . Then $r, r \in X_i$ so $(r, r) \in E$.

symmetry: let i be s.t. $r, s \in X_i$. Then $s, r \in X_i$ so $(s, r) \in E$.

transitivity: $(r, s) \in E$
 $(s, m) \in E$

$\Rightarrow \exists i_1$ and i_2 s.t. $r, s \in X_{i_1}$
 $s, m \in X_{i_2}$

~~$\{r, s\}$~~
 since the X_j 's are disjoint.

$i_1 = i_2 \Rightarrow r, m \in X_{i_1} = X_{i_2}$

so $(r, m) \in E$.